Research Highlights

Computerized Medical Imaging and Graphics

Highlights

- CAL-3DH unifies nested calibration and 3D hysteresis for glioma MRI segmentation.
- Controlled 5-fold audit shows random slice splitting inflates Dice by +8.79 pp.
- Nested temperature scaling (T*=1.844) cuts expected calibration error by 46.9%.
- Anisotropic 3D Z-hysteresis reduces tumor-free slice false alarms by 42.4%.
- 3D volumes reach ICC=0.8894 and preserve COCCluster radiogenomic signatures.